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SETH: Subcode Equivalence in the Head

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Digital Signature Scheme

- Public key cryptographic protocol for verifying authenticity of digital messages and/or documents
- Produces a signature bound to the message/document
- Produces a signature that is unforgeable
- Three algorithms:
  1. $\mathrm{KeyGen}(1^\lambda) \xrightarrow{\$} (\mathrm{pk}, \mathrm{sk})$
  2. $\mathrm{Sign}(\mathrm{m}, \mathrm{sk}) \rightarrow$ signature σ for the message m
  3. $\mathrm{Verify}(\mathrm{m}, \sigma, \mathrm{pk}) \rightarrow$ accept or reject

Post-Quantum Signatures from Code-Based Assumptions

- 2022: NIST (National Institute of Standards and Technology) announces post-quantum signature standards $\rightarrow$ only lattice-based schemes
- 2023: Call for additional digital signature schemes
- 2024: Round-2 candidates are announced, 5 code-based schemes

SDitH
CROSS
LESS
PERK
RYDE

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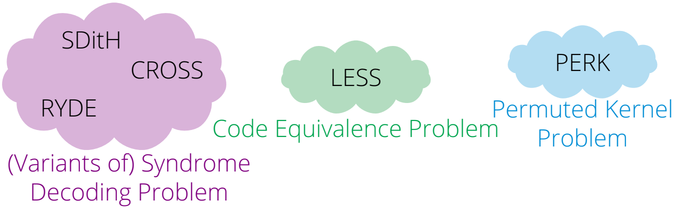

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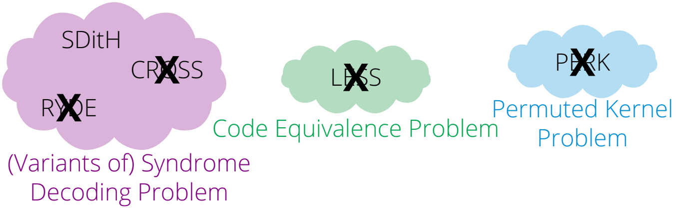

Why was LESS eliminated?

LESS = Linear Equivalence Signature Scheme

Linear Equivalence Problem (LEP)

Given $\mathscr{C}, \mathscr{C}' \subseteq \mathbb{F}_q^n$ be two codes of dimension k, with generator matrices $\mathbf{G}, \mathbf{G}' \in \mathbb{F}_q^{k\times n}$ respectively. Find, if they exist, a monomial matrix $\mathbf{Q} \in \mathcal{M}_n(q)$ and $\mathbf{S} \in GL_k(q)$ such that $\mathbf{G}' = \mathbf{SGQ}$.

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Public key = $(\mathbf{G}, \mathbf{G}')$
Secret key = $(\mathbf{S}, \mathbf{Q})$

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Public key = $(\mathbf{G}, \mathbf{G}')$
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Major deal breaker: **public keys are too large**!

Sizes for category 1 parameters ($q = 127$, $n = 252$, $k = 126$):

- Signature = 2625 B / 1329 B
- Public key = 13940 B / 97484 B

How can we obtain smaller public keys for code equivalence?

- Target more efficient representation of codes

How can we obtain smaller public keys for code equivalence?

- Target more efficient representation of codes
- **Slightly change the underlying problem**

Subcode Equivalence Problem

Subcode Equivalence Problem (SEP)

Given $\mathscr{C}, \mathscr{C}' \subseteq \mathbb{F}_q^n$ be two codes with respective dimension k and $k' < k$, find, if it exists, a permutation $\pi \in \mathrm{S}_n$ such that $\pi(\mathscr{C}') \subset \mathscr{C}$ (i.e. $\pi(\mathscr{C}')$ is a subcode of $\mathscr{C}$).

Subcode Equivalence Problem

Subcode Equivalence Problem (SEP)

Given $\mathscr{C}$ with generator matrix $\mathbf{G} \in \mathbb{F}_q^{k \times n}$ and $\mathcal{C}'$ with generator matrix $\mathbf{G}' \in \mathbb{F}_q^{k' \times n}$, with $k' < k$, find, if they exist, a full-rank matrix $\mathbf{S} \in \mathbb{F}_q^{k' \times k}$ and a permutation matrix $\mathbf{P} \in \mathrm{S}_n$ such that $\mathbf{G}' = \mathbf{SGP}$.

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Permutation Equivalence Problem (PEP)

Given two codes $\mathcal{C}, \mathcal{C}' \subset \mathbb{F}_q^n$ with generator matrices $\mathbf{G}, \mathbf{G}' \in \mathbb{F}_q^{k \times n}$, respectively, find, if they exist, an invertible matrix $\mathbf{S} \in \mathrm{GL}_k(q)$ and a permutation matrix $\mathbf{P} \in \mathrm{S}_n$ such that $\mathbf{G}' = \mathbf{SGP}$.

Why SEP?

- Very similar to PEP (and LEP)
- Public keys are indeed smaller
  - $\mathrm{pk}_{\mathrm{PEP}} = (\mathbf{G}, \mathbf{G}') \in \mathbb{F}_q^{k \times n} \times \mathbb{F}_q^{k \times n}$
  - $\mathrm{pk}_{\mathrm{SEP}} = (\mathbf{G}, \mathbf{G}') \in \mathbb{F}_q^{k \times n} \times \mathbb{F}_q^{k' \times n}$, with $k' < k$

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- Very similar to PEP (and LEP)
- Public keys are indeed smaller
- SEP is NP-complete [7], while PEP (and also LEP) is likely NOT NP-complete (or the polynomial hierarchy collapses)
- Smaller parameters for the codes, for instance
  - LESS: $q = 127, n = 252, k = 126$
  - this work: $q = 2^4, n = 74, k = 36, (k' = 3)$ (and ∃ secure instances even for $q = 2$)

$\Longrightarrow$ even smaller public keys ☺

Why SEP?

- Very similar to PEP (and LEP)
- Public keys are indeed smaller
- SEP is NP-complete [7], while PEP (and also LEP) is likely NOT NP-complete (or the polynomial hierarchy collapses)
- Smaller parameters for the codes
- SEP has fewer applications in cryptography
  - 2017: identification protocol based on SEP + syndrome decoding problem [7]
  - 2026: signature scheme based on subcode equivalence of matrix codes (rank metric) [4]

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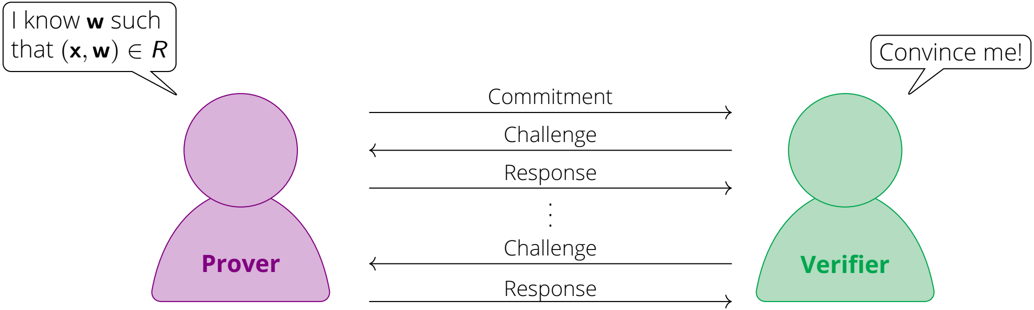
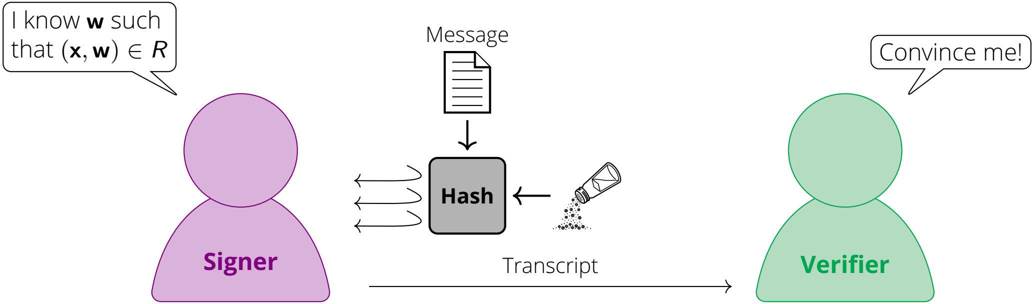
Goal: Design a signature scheme based on SEP (in the Hamming metric)

Identification Protocols

Given a hard relation R and a public statement $\mathbf{x}$...



Signatures from Identification Protocols: Fiat-Shamir Transform



$\rightarrow$ **Transcript**: List of messages produced by the signer

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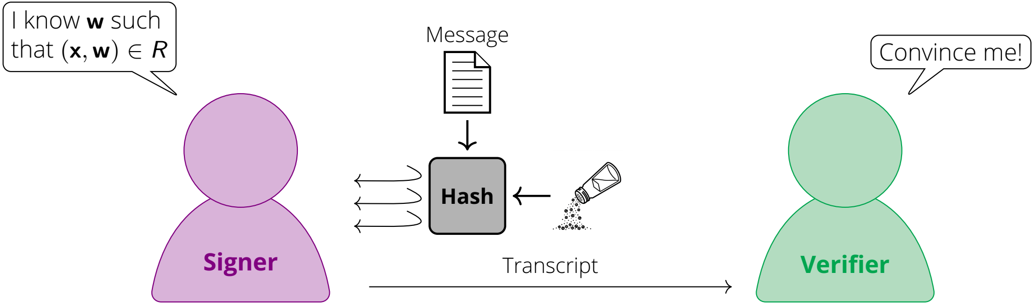

- **Correctness**: If the prover is honest, the verifier will always be convinced.
- **Soundness**: A dishonest prover can convince a verifier of a false statement with at most a negligible probability.
- **Zero-knowledge**: The verifier learns nothing about the prover's secret except for its validity.

Can we use LESS' identification scheme?

- R = PEP
- $\mathbf{x} = \mathrm{pk} = (\mathbf{G}', \mathbf{G}) \in \mathbb{F}_q^{k \times n} \times \mathbb{F}_q^{k \times n}$ generator matrices of $\mathcal{C}', \mathcal{C} \subset \mathbb{F}_q^n$
- $\mathbf{w} = \mathrm{sk} = (\mathbf{S}, \mathbf{P}) \in \mathrm{GL}_k(q) \times \mathrm{S}_n$ such that $\mathbf{G}' = \mathbf{SGQ}$

Prover		Verifier
$\tilde{\mathbf{P}} \xleftarrow{\$} \mathrm{S}_n$		
$h = \mathrm{Hash}(\mathrm{SF}(\mathbf{G}\tilde{\mathbf{P}}))$	$\xrightarrow{\quad h \quad}$	
	$\xleftarrow{\quad b \quad}$	$b \xleftarrow{\$} \{0, 1\}$
If $b = 0$: $\mathrm{rsp} = \tilde{\mathbf{P}}$		
If $b = 1$: $\mathrm{rsp} = \mathbf{P}^{-1}\tilde{\mathbf{P}}$	$\xrightarrow{\quad \mathrm{rsp} \quad}$	
		If $b = 0$: $\mathrm{Hash}(\mathrm{SF}(\mathbf{G} \cdot \mathrm{rsp})) \stackrel{?}{=} h$
		If $b = 1$: $\mathrm{Hash}(\mathrm{SF}(\mathbf{G}' \cdot \mathrm{rsp})) \stackrel{?}{=} h$

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- R = SEP
- $\mathbf{x} = (\mathbf{G}', \mathbf{G}) \in \mathbb{F}_q^{k' \times n} \times \mathbb{F}_q^{k \times n}$ generator matrices of $\mathcal{C}', \mathcal{C} \subset \mathbb{F}_q^n$
- $\mathbf{w} = (\mathbf{S}, \mathbf{P}) \in \mathbb{F}_q^{k' \times k} \times \mathrm{S}_n$, $\mathrm{rk}(\mathbf{S}) = k'$, such that $\mathbf{G}' = \mathbf{SGP}$

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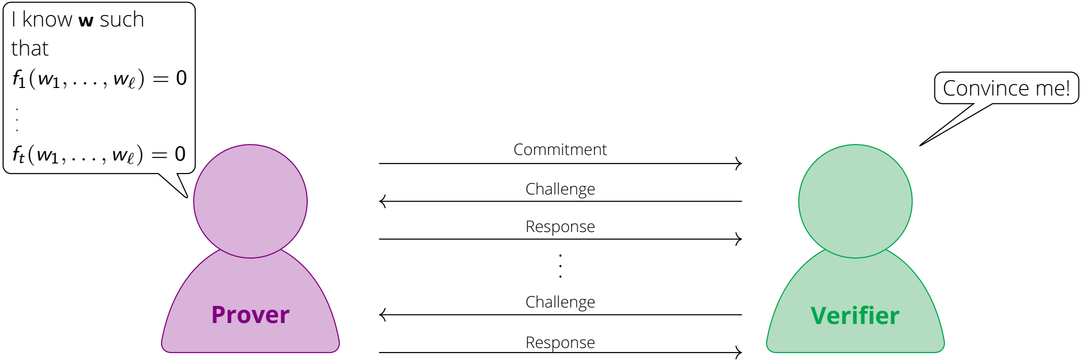
Problem: $\dim(\mathbf{G}\tilde{\mathbf{P}}) \neq \dim(\mathbf{G}'\mathbf{P}^{-1}\tilde{\mathbf{P}}) \implies \mathrm{Hash}(\mathrm{SF}(\mathbf{G}' \cdot \mathrm{rsp})) \neq h$

Polynomial Interactive Oracle Proof (PIOP)



Polynomial Interactive Oracle Proof (PIOP)

Prover		Verifier
for $i = 1, \ldots, \ell$: $P_i(X) = w_i X + v_i,\ v_i \xleftarrow{\$} \mathbb{F}_{q^r}$ for $i = 1, \ldots, t$: $F_i(X) = X^d \cdot f_i\left(\frac{P_1(X)}{X}, \ldots, \frac{P_\ell(X)}{X}\right)$	$\xrightarrow{\text{Commit}}$	
	$\xleftarrow{\gamma_1, \ldots, \gamma_t}$	$\gamma_1, \ldots, \gamma_t \xleftarrow{\$} \mathbb{F}_{q^r}$
$G(X) = \sum_{i=1}^{t} \gamma_i F_i(X)$	$\xrightarrow{G(X)}$	
	$\xleftarrow{\Delta}$	$\Delta \xleftarrow{\$} \mathbb{F}_{q^r}$
$s_j = P_j(\Delta)$	$\xrightarrow[\text{Opening}]{s_1, \ldots, s_\ell}$	$\deg(G(X)) \stackrel{?}{=} d - 1$ $G(\Delta) \stackrel{?}{=} \sum_{i=1}^{t} F_i(\Delta)$

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Correctness

$$F_i(X) = \underbrace{f_i(w_1, \dots, w_\ell)}_{=0} X^d + \dots$$

Polynomial Interactive Oracle Proof (PIOP)

Prover

for $i = 1, \dots, \ell$:

$P_i(X) = w_i X + v_i, \ v_i \xleftarrow{\$} \mathbb{F}_{q^r}$

for $i = 1, \dots, t$:

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$\xrightarrow{\text{Commit}}$

Verifier

$\gamma_1, \dots, \gamma_t \xleftarrow{\$} \mathbb{F}_{q^r}$

$\xleftarrow{\gamma_1, \dots, \gamma_t}$

$G(X) = \sum_{i=1}^t \gamma_i F_i(X) + Q(X)$

$\xrightarrow{G(X)}$

$\Delta \xleftarrow{\$} \mathbb{F}_{q^r}$

$\xleftarrow{\Delta}$

$s_j = P_j(\Delta)$

$\tilde{s} = Q(\Delta)$

$\xrightarrow[\text{Opening}]{s_1, \dots, s_\ell, \tilde{s}}$

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Zero knowledge

Add mask $Q(X)$ of degree $d - 1$

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Commit $\longrightarrow$

$\xleftarrow{\gamma_1, \ldots, \gamma_t}$

$\xrightarrow{G(X)}$

$\xleftarrow{\Delta}$

$\xrightarrow{s_1, \ldots, s_\ell, \tilde{s}}$ Opening

Verifier

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Zero knowledge

Add mask $Q(X)$ of degree $d - 1$

Soundness

$\leq \frac{d}{q^r}$

Flavours of PIOP: TCitH and VOLEitH

Concrete instances of PIOP: Threshold Computation in the Head and VOLE in the Head

- Generation and commitment of degree-1 polynomials: batch vector commitments via GGM trees
- Different soundness amplification techniques:
  - **TCitH**: multiple polynomials $G(X)$ and parallel repetitions
  - **VOLEitH**: lift $P_j(X) = w_j X + v_j$ to exponentially large field
- VOLEitH signatures are generally shorter

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N.B.: problem-agnostic approach

Multivariate Polynomials from SEP[1]

Let $\mathcal{C}, \mathcal{C}' \subseteq \mathbb{F}_q^n$ be two subcode equivalent code of dimension k and $k' < k$, respectively. Let $\mathbf{H} \in \mathbb{F}_q^{(n-k)\times n}$ be a parity check matrix of $\mathcal{C}$ and $\mathbf{G}' \in \mathbb{F}_q^{k\times n}$ a generator matrix $\mathcal{C}'$. Then, if $\mathbf{P}$ is the permutation matrix associated to $\pi \in S_n$ such that $\pi(\mathcal{C}') \subset \mathcal{C}$, then

$$\mathbf{G}'\mathbf{P}\mathbf{H}^\top = \mathbf{0}$$

Goal: Given $\mathbf{G}'$ and $\mathbf{H}$, prove knowledge of $\mathbf{P}$ such that:

$\mathbf{G}'\mathbf{P}\mathbf{H}^\top = \mathbf{0}$	Each row of $\mathbf{P}$ has exactly one entry $= 1$	Each column of $\mathbf{P}$ has exactly one entry $= 1$

[1] Adapted from [8]

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Each row of $\mathbf{P}$ has exactly one entry $= 1$	Each column of $\mathbf{P}$ has exactly one entry $= 1$

$\mathbf{P}$ is a permutation matrix

[1] Adapted from [8]

Multivariate Polynomials from SEP

Let $d = \lceil \log_2(n) \rceil$. For $i = 0, \ldots, n-1$:

- pos_i position of the 1 in the i-th row of $\mathbf{P}$
- $\mathcal{B}^i \in \mathbb{F}_2^d$ binary expansion of i
- $\mathcal{B}^{\mathrm{pos}_i} = (p_{i,0}, \ldots, p_{i,d-1})$
- For $j = 0, \ldots, n-1$.

$$Z(i,j) := \prod_{l=0}^{d-1} \left(p_{i,l} \oplus \mathcal{B}^j{}_l \oplus 1 \right) = 1 \iff j = \mathrm{pos}_i$$

$\implies (Z(i,0), \ldots, Z(i,n-1))$ is the i-th row of $\mathbf{P}$

Multivariate Polynomials from SEP

For $i, j = 0, \dots, n-1,\ Z(i,j) := \prod_{l=0}^{d-1} \left(p_{i,l} \oplus \mathcal{B}^j{}_l \oplus 1\right)$:

- Variables $(p_{i,j})_{i,j=0,\dots,n-1}$ represent a permutation matrix **P**:

$$\mathcal{S}_{\text{col}} = \left\{ \left(\sum_{i=0}^{n-1} Z(i,j) \right) - 1 = 0 \mid j \in [0, n-1] \right\}$$

- **P** realizes the subcode equivalence:

$$\mathcal{S}_{\text{eq}} = \left\{ \sum_{m=0}^{n-1} \left(\mathbf{G}'_{i,m} \cdot \sum_{l=0}^{n-1} Z(m,l) \cdot \mathbf{H}j, l \right) = 0 \mid i \in [0, k'-1], j \in [0, n-k-1] \right\}$$

Multivariate Polynomials from SEP

For $i, j = 0, \ldots, n-1,\ Z(i,j) := \prod_{l=0}^{d-1} \left(p_{i,l} \oplus \mathcal{B}^j{}_l \oplus 1 \right)$:

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$$\mathcal{S}_{\mathrm{eq}} = \left\{ \sum_{m=0}^{n-1} \left(\mathbf{G}'_{i,m} \cdot \sum_{l=0}^{n-1} Z(m,l) \cdot \mathbf{H}j, l \right) = 0 \mid i \in [0, k'-1], j \in [0, n-k-1] \right\}$$

$\rightarrow k' \cdot (n-k) + n$ equations of degree at most $d = \lceil \log_2(n) \rceil$ in $n \cdot d$ variables
$\rightarrow$ works only for $q = 2^m, m > 0$

Size Comparison

Code parameters for this work: $(q, n, k', k) = (2^4, 74, 3, 36)$

Scheme	Signature Size (B)	Public Key (B)
This work (VOLE)	3986	106
This work (TCitH)	5287	106
VOLE PKP	3600	100
PERK (small)	3480	100
PERK (fast)	4320	100
LESS 252-192	2289	13940
LESS 252-45	1329	97484
LESS VOLE	49877	13940
Matrix Subcode	4788	255

Summary and Future Work Directions

To summarize...

- Signatures through Polynomial Interactive Proofs (PIOPs) based on SEP
- Multivariate polynomial system from SEP
- Comparison with existing schemes

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- Multivariate polynomial system from SEP
- Comparison with existing schemes

To do's:

- Implementation of the signature scheme

References I

[1] Gorjan Alagic et al. "Status Report on the Second Round of the Additional Digital Signature Schemes for the NIST Post-Quantum Cryptography Standardization Process". In: NIST IR 8610 (2026). url: `https://doi.org/10.6028/NIST.IR.8610`.

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Thank you for the attention!
Questions?

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